

3.3.1 Investigating biological problems involves changing a specific factor, the independent variable, and measuring the changes in the dependent variable that result.

Candidates should be able to

- use knowledge and understanding from the AS specification to pose scientific questions and define scientific problems
- identify the independent variable and describe an appropriate method of varying it in such detail that a student starting an AS course could carry out the suggested procedure without further assistance
- identify other variables that might be expected to exert a significant influence on the results, use knowledge from relevant parts of the AS specification to explain why, and describe how these would be kept constant
- where necessary, describe how and explain why appropriate control experiments should be established
- identify the dependent variable and describe how they would collect a full range of useful quantitative data, measured to an appropriate level of accuracy and precision
- distinguish between accuracy and reliability and describe precautions needed to obtain valid, accurate and reliable data.

Practical work carried out in the context of Units 1 and 2 should enable candidates to gain experience of

- the use of water baths to change or control temperature
- the use of buffers to change or control pH
- producing an appropriate dilution series when provided with stock solutions of reagents.